

NAME:

Practice Problems for Week 3

Please complete, bring to class (written answers) and submit on gradescope (code) by end of day on **Monday 4/20/26**.

You will be submitting the following files:

- birthday.py
- cricket.py

Programming Part 1: Happy Birthday

1. Download the starter file `birthday.py` from the course website and read the instructions inside before continuing.
2. Write a function that “sings” Happy Birthday for someone. The function should take the person’s name as a parameter and then print Happy Birthday in the following format.:

```
Happy Birthday to you!  
Happy Birthday to you!  
Happy Birthday dear <name>!  
Happy Birthday to you!
```

3. Make sure to include appropriate comments.
4. Save your program using the filename `birthday.py`.

Programming Part 2: Cricket Chirping

1. Download the starter file `cricket.py` from the course website and read the instructions inside before continuing.
2. Amos Dolbaer discovered that, for certain types of crickets, there’s a relationship between the temperature and the rate at which they chirp (https://en.wikipedia.org/wiki/Dolbear's_law). More specifically, the relationship is as follows:

$$T = 50 + (N - 40)/4$$

where T is the temperature in Farenheight and N the number of chirps per minute.

Write a function that calculates the temperature when given the number of times a cricket chirps in 15 seconds. Your function should do the following:

- a. calculate the chirps per minute
 - b. calculate the temperature based on chirps per minute
 - c. return the temperature
3. Save your program in a file called `cricket.py`

Written Questions

1. 1. Which of the following are valid first lines for a Python function definition? Please explain why the invalid ones are not valid.

a. `def plus ten (anumber):`

b. `def double (anumber)`

c. `def greeting:`

d. `percentage(total, part):`

e. `def print_introduction():`

f. `def move()`

g. `def goto(x coordinate y coordinate):`

3. Print or return?

Consider the following (incomplete) function definitions.

Given the way these functions are used in the code at the end, complete the dotted lines by either turning them into a statement that **returns** the indicated value or into a statement that **prints** the indicated value.

Also complete the dotted lines in the code at the end by choosing to either **return** or **print** the value of the variable `a`.

```
def f1():  
    a = int(input('Please input a number: '))  
  
    .....a.....
```

```
def f2(var):  
    squared = var**2  
  
    .....squared.....
```

```
def f3(var):  
  
    .....var**3.....
```

```
a = 10  
  
.....a.....  
f2(a)  
print(f1() + 3)  
a = f3(a) * 5  
  
.....a.....
```

f. Assuming that the user input 1 when asked for a number, what will this code print to the screen?

```
.....  
.....  
.....  
.....  
.....  
.....
```

4. Check-in

4.a. Were you able to complete all of the programming activities (yes/no)?

.....

4.b. What, if anything, did you struggle with when working on the activities?

.....

.....

.....

.....

4.c. If you managed to overcome the challenges, how did you do it?

.....

.....

.....

.....